

TraitleZ-Data

Mohid Latif

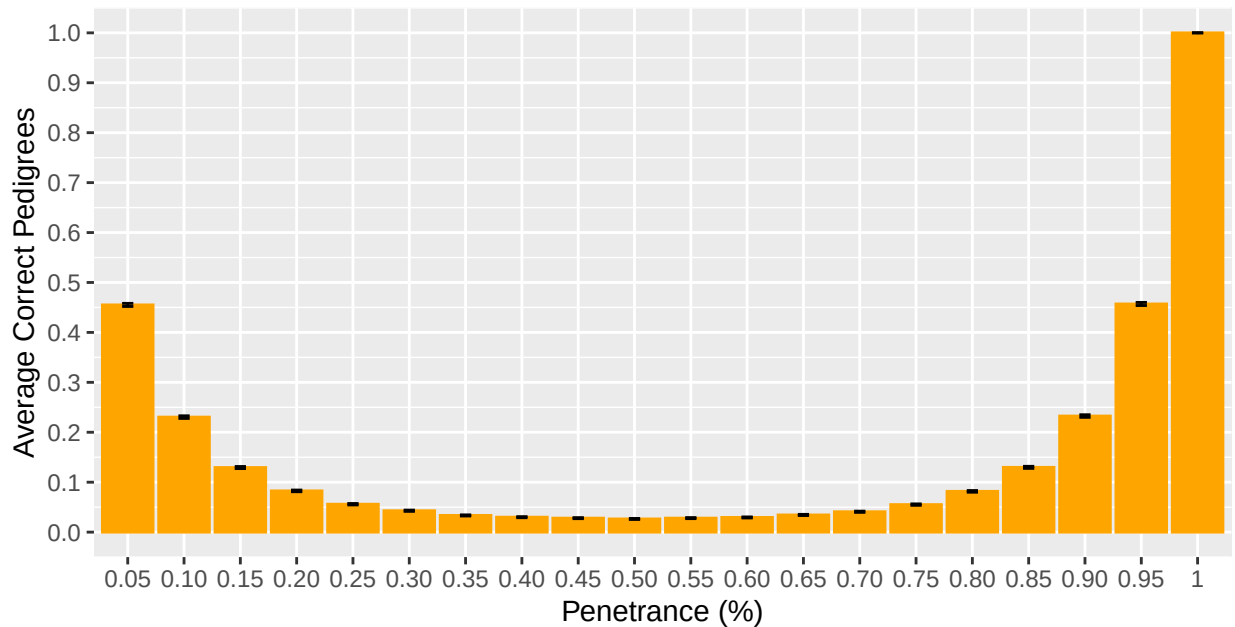
2026-03-18

```
library("ggplot2")
library("plotrix")
library("stringr")
all.data = read.csv("results.csv")
diseaseType = all.data$DiseaseType
penetrance = all.data$Penetrance
correct = as.integer(all.data$Correct=="true")
cnt = 0
M.corrected = c()
M.sd = c()
start = 1
end = 100000
for(i in 1:20){
  M.corrected = c(M.corrected, mean(correct[start:end]))
  M.sd = c(M.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}

dfM <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60", "0.55", "0.50", "0.45", "0.40", "0.35", "0.30", "0.25", "0.20", "0.15", "0.10", "0.05", "0.00"))
ggplot(dfM, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "orange", color = "orange") +
  labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Mitochondrial Pedigrees with different levels of Penetrance"),
    subtitle = str_wrap("Traitle Z"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 and F2 generations")
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
  ) +
  geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2) +
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```

Classification Accuracy for Mitochondrial Pedigrees with different levels of Penetrance

Traitle Z



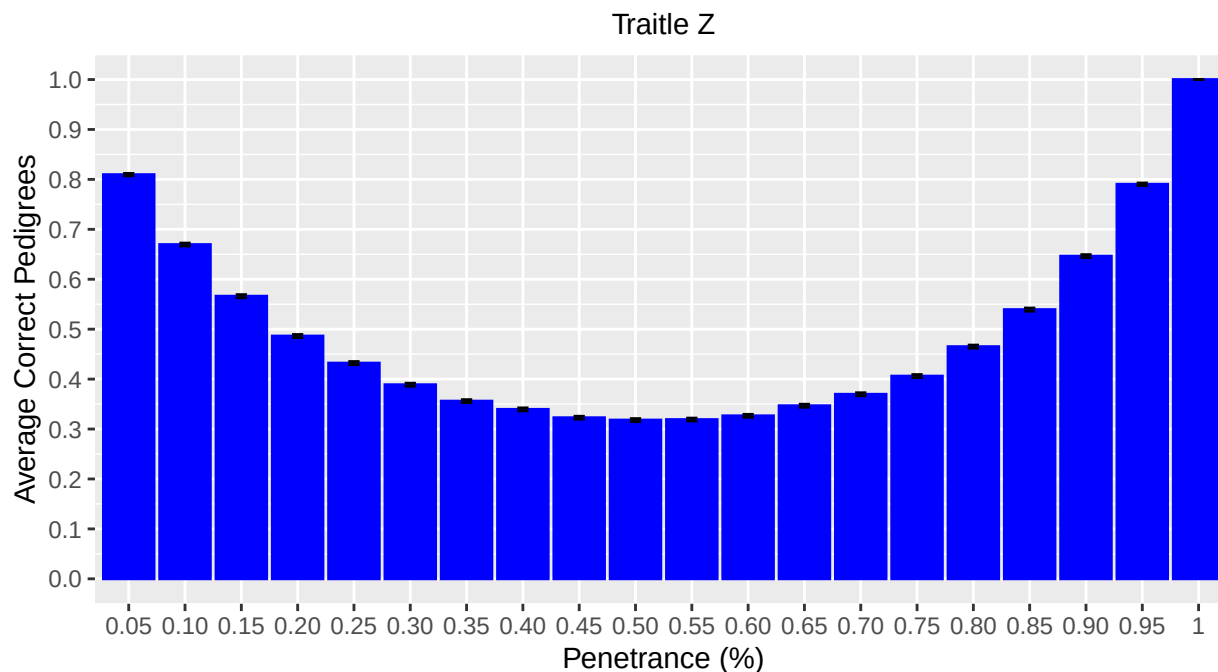
The parental generation consists of 2 individuals. In the F1 and F2 generations, each parent has 1–3 children, and in the F3 generation, each F2 individual has 2–3 children.

```
XLD.corrected = c()
XLD.sd = c()
for(i in 1:20){
  XLD.corrected = c(XLD.corrected, mean(correct[start:end]))
  XLD.sd = c(XLD.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}

dfXLD <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60",
ggplot(dfXLD, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "blue", color = "blue") +
  labs(
    x = str_wrap("Penetrance (%)" ),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for X-Linked Dominant Pedigrees with different levels of I",
    subtitle = str_wrap("Traitle Z"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 and F2 generations
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
  ) +
  geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2) +
```

```
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```

Classification Accuracy for X-Linked Dominant Pedigrees with different levels Penetrance

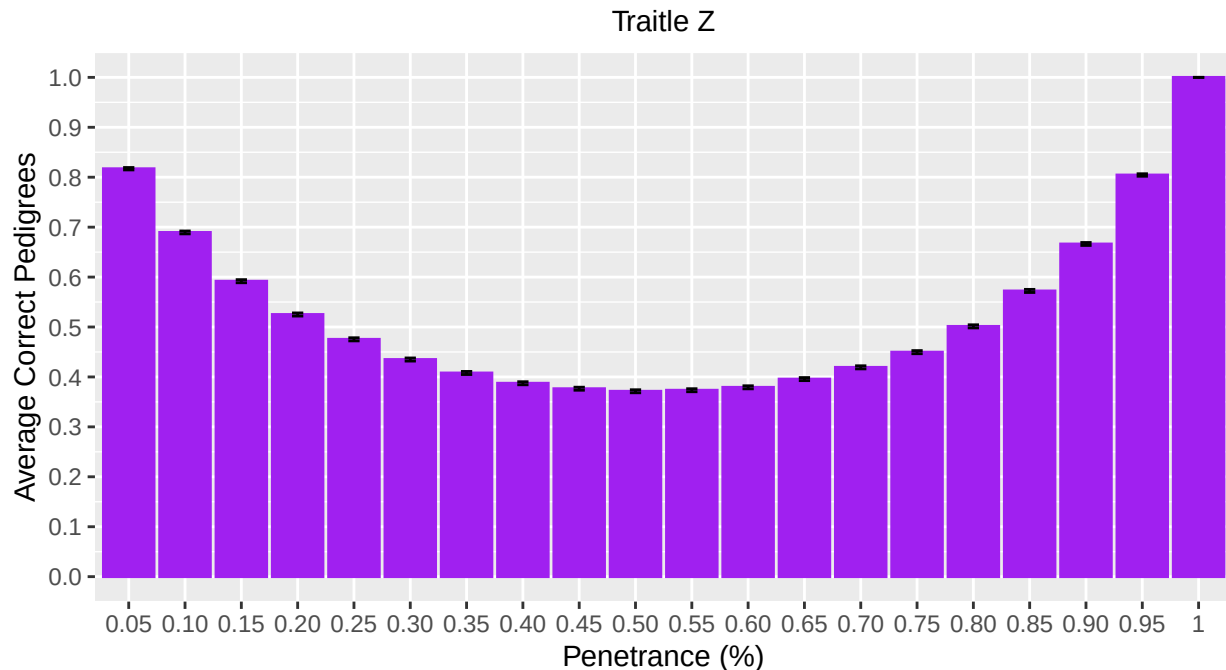


The parental generation consists of 2 individuals. In the F1 and F2 generations, each parent has 1–3 children, and in the F3 generation, each F2 individual has 2–3 children.

```
XLR.corrected = c()
XLR.sd = c()
for(i in 1:20){
  XLR.corrected = c(XLR.corrected, mean(correct[start:end]))
  XLR.sd = c(XLR.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}
dfXLR <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60",
ggplot(dfXLR, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "purple", color = "purple") +
  labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for X-Linked Recessive Pedigrees with different levels of
    subtitle = str_wrap("Traitle Z"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 and F2 generations
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
  ) +
```

```
geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2) +
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```

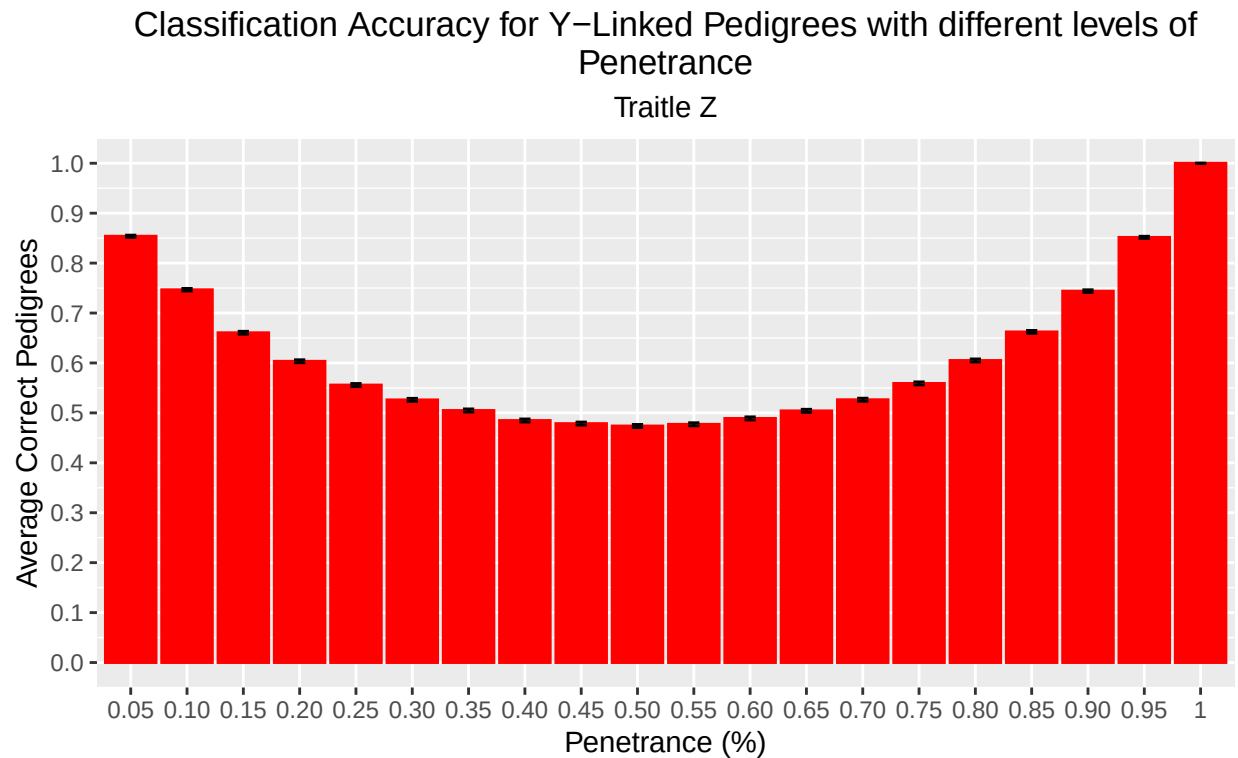
Classification Accuracy for X-Linked Recessive Pedigrees with different level: of Penetrance



The parental generation consists of 2 individuals. In the F1 and F2 generations,
each parent has 1–3 children, and in the F3 generation, each F2 individual has
2–3 children.

```
YL.corrected = c()
YL.sd = c()
for(i in 1:20){
  YL.corrected = c(YL.corrected, mean(correct[start:end]))
  YL.sd = c(YL.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}
dfYL <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60", "0.55", "0.50", "0.45", "0.40", "0.35", "0.30", "0.25", "0.20", "0.15", "0.10", "0.05"))
ggplot(dfYL, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "red", color = "red") +
  labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Y-Linked Pedigrees with different levels of Penetrance"),
    subtitle = str_wrap("Traitle Z"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 and F2 generations, each parent has 1–3 children, and in the F3 generation, each F2 individual has 2–3 children.")
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
```

```
)+
geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))
```



The parental generation consists of 2 individuals. In the F1 and F2 generations, each parent has 1–3 children, and in the F3 generation, each F2 individual has 2–3 children.

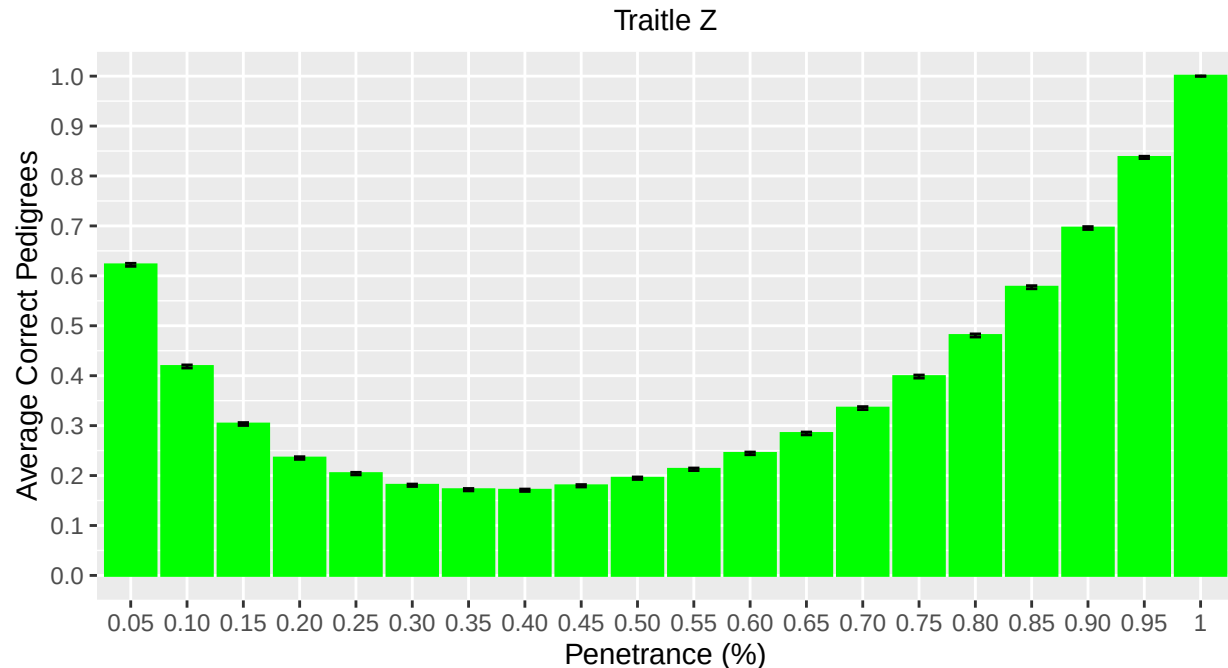
```
AD.corrected = c()
AD.sd = c()
for(i in 1:20){
  AD.corrected = c(AD.corrected, mean(correct[start:end]))
  AD.sd = c(AD.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}
dfAD <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60", "0.55", "0.50", "0.45", "0.40", "0.35", "0.30", "0.25", "0.20", "0.15", "0.10", "0.05"))
ggplot(dfAD, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "green", color = "green") +
  labs(
    x = str_wrap("Penetrance (%)" ),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Autosomal Dominant Pedigrees with different levels of Penetrance"),
    subtitle = str_wrap("Traitle Z"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 and F2 generations, each parent has 1–3 children, and in the F3 generation, each F2 individual has 2–3 children.")
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(hjust = 0.5),
    plot.caption = element_text(hjust = 0.5)
```

```

plot.caption = element_text(hjust = 0.5)
)+
geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))

```

Classification Accuracy for Autosomal Dominant Pedigrees with different level of Penetrance



The parental generation consists of 2 individuals. In the F1 and F2 generations, each parent has 1–3 children, and in the F3 generation, each F2 individual has 2–3 children.

```

AR.corrected = c()
AR.sd = c()
for(i in 1:20){
  AR.corrected = c(AR.corrected, mean(correct[start:end]))
  AR.sd = c(AR.sd, 1.96*std.error(correct[start:end]))
  start = start + 100000
  end = end + 100000
}
dfAR <- data.frame(Penetrance = c("1", "0.95", "0.90", "0.85", "0.80", "0.75", "0.70", "0.65", "0.60", "0.55", "0.50", "0.45", "0.40", "0.35", "0.30", "0.25", "0.20", "0.15", "0.10", "0.05"))
ggplot(dfAR, aes(x = Penetrance, y = AverageCorrect)) +
  geom_bar(stat = "identity", fill = "pink", color = "pink") +
  labs(
    x = str_wrap("Penetrance (%)"),
    y = str_wrap("Average Correct Pedigrees"),
    title = str_wrap("Classification Accuracy for Autosomal Recessive Pedigrees with different levels of Penetrance"),
    subtitle = str_wrap("Traitle Z"),
    caption = str_wrap("The parental generation consists of 2 individuals. In the F1 and F2 generations, each parent has 1–3 children, and in the F3 generation, each F2 individual has 2–3 children.")
  ) +
  theme(
    plot.title = element_text(hjust = 0.5),

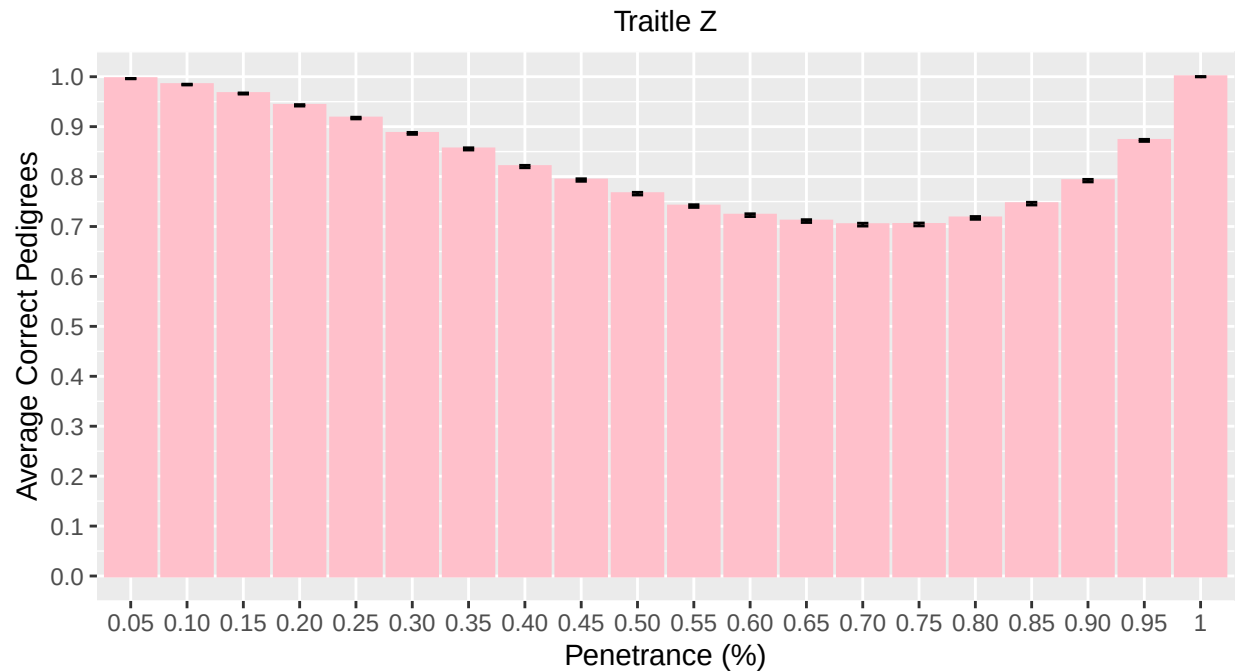
```

```

plot.subtitle = element_text(hjust = 0.5),
plot.caption = element_text(hjust = 0.5)
)+
geom_errorbar(aes(ymin = AverageCorrect - sd, ymax = AverageCorrect + sd), width = 0.2)+
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, by = 0.1))

```

Classification Accuracy for Autosomal Recessive Pedigrees with different level of Penetrance



The parental generation consists of 2 individuals. In the F1 and F2 generations, each parent has 1–3 children, and in the F3 generation, each F2 individual has 2–3 children.

```

all.data$Correct = as.numeric(as.logical(all.data$Correct))

logistic.all <- glm(Correct ~ DiseaseType * (Penetrance + I(Penetrance^2)), data=all.data, family="binomial")
summary(logistic.all)

```

```

##
## Call:
## glm(formula = Correct ~ DiseaseType * (Penetrance + I(Penetrance^2)),
##     family = "binomial", data = all.data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.725499   0.005051  143.622  <2e-16 ***
## DiseaseTypeAR    4.833497   0.014184  340.762  <2e-16 ***
## DiseaseTypeM     0.120732   0.007881   15.319  <2e-16 ***
## DiseaseTypeXLD    1.100529   0.007316  150.422  <2e-16 ***
## DiseaseTypeXLR    1.101023   0.007343  149.947  <2e-16 ***
## DiseaseTypeYL     1.262535   0.007531  167.637  <2e-16 ***
## Penetrance     -11.008662   0.024520 -448.973  <2e-16 ***

```

```
## I(Penetrance^2)          12.712388    0.024135   526.720    <2e-16 ***
## DiseaseTypeAR:Penetrance -3.871561    0.052537   -73.692    <2e-16 ***
## DiseaseTypeM:Penetrance -10.001925   0.045599  -219.346    <2e-16 ***
## DiseaseTypeXLD:Penetrance -0.074745   0.034143    -2.189    0.0286 *
## DiseaseTypeXLR:Penetrance  0.868547    0.034065    25.497    <2e-16 ***
## DiseaseTypeYL:Penetrance  1.898480    0.034623    54.833    <2e-16 ***
## DiseaseTypeAR:I(Penetrance^2) -1.017860   0.045245   -22.496    <2e-16 ***
## DiseaseTypeM:I(Penetrance^2)  8.771791    0.044074   199.024    <2e-16 ***
## DiseaseTypeXLD:I(Penetrance^2) -1.566782   0.032981   -47.506    <2e-16 ***
## DiseaseTypeXLR:I(Penetrance^2) -2.493320   0.032897   -75.791    <2e-16 ***
## DiseaseTypeYL:I(Penetrance^2) -3.413348   0.033487  -101.931    <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 16634928  on 11999999  degrees of freedom
## Residual deviance: 12450945  on 11999982  degrees of freedom
## AIC: 12450981
##
## Number of Fisher Scoring iterations: 6
```

```
M.log <- all.data %>%
  dplyr::filter(DiseaseType=="M")
logistic.M <- glm(Correct ~ Penetrance + I(Penetrance^2), data=M.log, family="binomial")
summary(logistic.M)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
## data = M.log)
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.846231  0.006049  139.9    <2e-16 ***
## Penetrance   -21.010587  0.038445  -546.5    <2e-16 ***
## I(Penetrance^2) 21.484179  0.036879  582.6    <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1759475  on 1999999  degrees of freedom
## Residual deviance: 1171242  on 1999997  degrees of freedom
## AIC: 1171248
##
## Number of Fisher Scoring iterations: 6
```

```
XLD.log <- all.data %>%
  dplyr::filter(DiseaseType=="XLD")
logistic.XLD <- glm(Correct ~ Penetrance + I(Penetrance^2), data=XLD.log, family="binomial")
summary(logistic.XLD)
```



```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##      data = XLD.log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.826028   0.005293   345.0   <2e-16 ***
## Penetrance     -11.083406   0.023759  -466.5   <2e-16 ***
## I(Penetrance^2)  11.145606   0.022477   495.9   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2772377  on 1999999  degrees of freedom
## Residual deviance: 2473883  on 1999997  degrees of freedom
## AIC: 2473889
##
## Number of Fisher Scoring iterations: 4
```

```
XLR.log <- all.data %>%
  dplyr::filter(DiseaseType=="XLR")
logistic.XLR <- glm(Correct ~ Penetrance + I(Penetrance^2), data=XLR.log, family="binomial")
summary(logistic.XLR)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##      data = XLR.log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.826522   0.005329   342.7   <2e-16 ***
## Penetrance     -10.140115   0.023647  -428.8   <2e-16 ***
## I(Penetrance^2)  10.219068   0.022355   457.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2764442  on 1999999  degrees of freedom
## Residual deviance: 2513425  on 1999997  degrees of freedom
## AIC: 2513431
##
## Number of Fisher Scoring iterations: 4
```

```
YL.log <- all.data %>%
  dplyr::filter(DiseaseType=="YL")
logistic.YL <- glm(Correct ~ Penetrance + I(Penetrance^2), data=YL.log, family="binomial")
summary(logistic.YL)
```

```
##
```

```
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##      data = YL.log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.988034   0.005586   355.9  <2e-16 ***
## Penetrance     -9.110182   0.024445  -372.7  <2e-16 ***
## I(Penetrance^2)  9.299040   0.023214   400.6  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2665254  on 1999999  degrees of freedom
## Residual deviance: 2469749  on 1999997  degrees of freedom
## AIC: 2469755
##
## Number of Fisher Scoring iterations: 4
```

```
AD.log <- all.data %>%
  dplyr::filter(DiseaseType=="AD")
logistic.AD <- glm(Correct ~ Penetrance + I(Penetrance^2), data=AD.log, family="binomial")
summary(logistic.AD)
```

```
##
## Call:
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##      data = AD.log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.725499   0.005051   143.6  <2e-16 ***
## Penetrance     -11.008662   0.024520  -449.0  <2e-16 ***
## I(Penetrance^2) 12.712388   0.024135   526.7  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2670014  on 1999999  degrees of freedom
## Residual deviance: 2181383  on 1999997  degrees of freedom
## AIC: 2181389
##
## Number of Fisher Scoring iterations: 4
```

```
AR.log <- all.data %>%
  dplyr::filter(DiseaseType=="AR")
logistic.AR <- glm(Correct ~ Penetrance + I(Penetrance^2), data=AR.log, family="binomial")
summary(logistic.AR)
```

```
##
## Call:
```

```
## glm(formula = Correct ~ Penetrance + I(Penetrance^2), family = "binomial",
##     data = AR.log)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)    5.55900    0.01325   419.4  <2e-16 ***
## Penetrance    -14.88022    0.04646  -320.2  <2e-16 ***
## I(Penetrance^2) 11.69453    0.03827   305.6  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1811718  on 1999999  degrees of freedom
## Residual deviance: 1641263  on 1999997  degrees of freedom
## AIC: 1641269
##
## Number of Fisher Scoring iterations: 6
```